

Dual-Tone Horn

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This circuit generates two different frequencies and mixes these to produce a balanced ringing

tone, which can be used in burglar alarms or call bells.

The circuit uses a dual-timer IC LM556 (IC1) and penta-watt audio power IC TDA2003 (IC2) (refer Fig. 1).

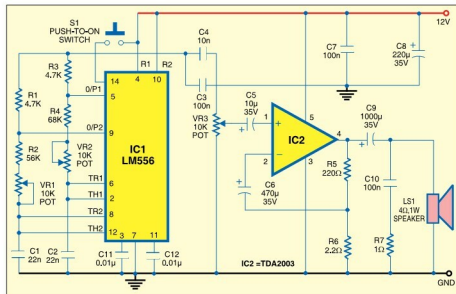


Fig. 1: Circuit for dual-tone horn



Both the timers of LM556 are wired as astable oscillators to generate square waves. The first timer is configured to generate a

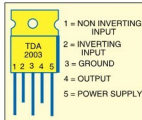


Fig. 2: Pin configuration of TDA2003

400Hz audible tone and the second timer for a 550Hz tone. The frequencies of both the oscillators are mixed and amplified by IC2. A higher frequency would result in shrill and unpleasant sound.

For better performance, the speaker should have a good-quality woofer with coil impedance of 4 ohms. Use a proper heat-sink for IC2. Adjust potentiometers VR1 and VR2 to achieve a pleasant sound. Potmeter VR3 is used for volume control.

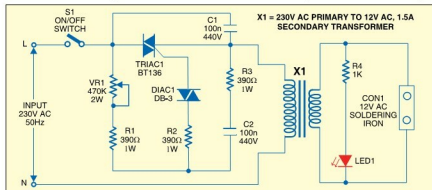
Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Refer Fig. 2 for pin configuration of TDA2003. Connect the switches at the front panel and the speaker at the back side of the cabinet. ●

Simple Soldering Iron Temperature Regulator

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Soldering irons are available in different wattages and usually run at 230V AC mains. However, these have no temperature control.

Low-voltage soldering irons (e.g., 12V) generally form part of a soldering station and are designed to be used with a temperature controller. A proper temperature-controlled soldering iron or station is expensive. Here is a simple



circuit that provides manual control of the temperature of an ordinary 12V AC soldering iron.

The circuit consists of power switch S1, TRIAC1, DIAC1, potentiometer VR1, resistors R1 to R3, capacitors C1 and C2, and step-down transformer X1. Adjusting the resistance of VR1 changes the charging rate of C1 to regulate the conduction angle of TRIAC1, and hence the output power (heat) of the low-voltage soldering iron connected to X1. The red LED (LED1) indicates the power status.

Assemble the circuit on a general-purpose PCB and enclose in a suitable plastic box. Since the front end of the circuit is directly connected to 230V AC mains supply, never attempt to operate the circuit without the cabinet. Use a heavy-duty potentiometer with plastic shaft and a knob for temperature control. ●